

3 Phase Induction Motor

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Table of contents

Project title	Error! Bookmark not defined.
Table of contents	1
1. Summary	1
2. Topic description	1
3. Modeling and simulation	2
4. Simulation: verification and prediction.....	2
5. Measurements	3
6. Comparison of Measurements to Simulations.....	8
7. Conclusion.....	9
Appendix.....	10

1. Summary

Our project consists of analyzing the performance of a three-phase induction motor and modeling its behavior. The process involved collecting data in both the Locked Rotor State, using a clamp, and Free Spinning state, allowing the motor to spin freely.

2. Topic description

This project models and experiments on a 1.5HP three phase induction motor. The core principle of the motor is to produce torque through the rotating magnetic field. The motor is rated to operate at 208V, and using the Variable Frequency Drive. We can harness the 120V AC power from the wall to drive the motor. The Variable Frequency Drive allows one to control frequency and voltage allowing us to perform tests at three different states (10, 20, 40 Hz).

Experiment Setup:

Equipment:

- Motor: 1.5HP, 3 Phase Induction Motor for 208V
- Variable Frequency Drive: Takes 120V input from wall, returns 3 phase 208V output
- Ratchet Straps: To tie down the motor while it spins
- Tachometer: Measures the rotational speed of the motor
- Multimeter: Measures the voltage and current of the line
- Scale: Measures weight to be used in calculations for force and then torque

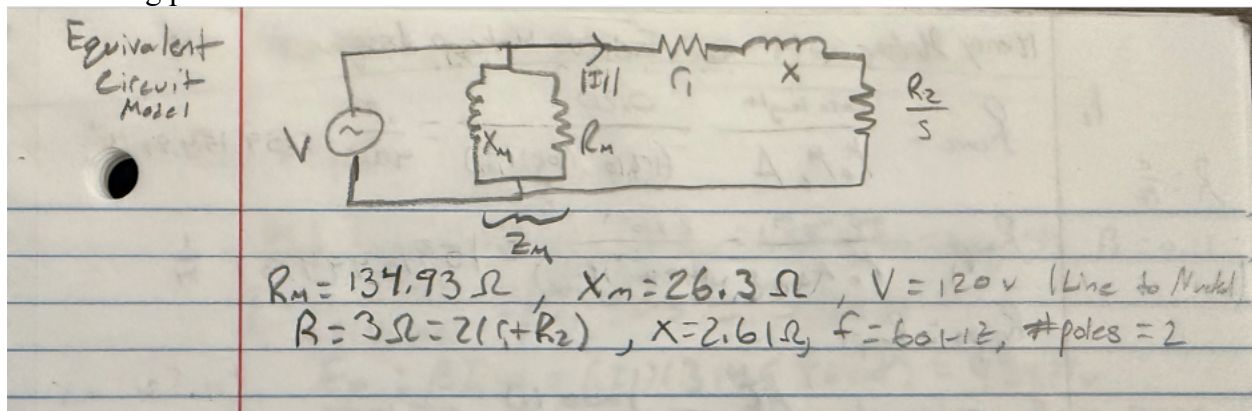
- Wrench: Clamps the motor for locked rotor measurements

3. Modeling and simulation

The transient behavior is modeled by the single phase circuit. This model includes:

- R (Stator Resistance)
- X (Stator Reactance)
- R_m (Magnetization Resistance)
- X_m (Magnetization Reactance)
- X (Rotor Reactance)
- R_2/s (Load Resistance)

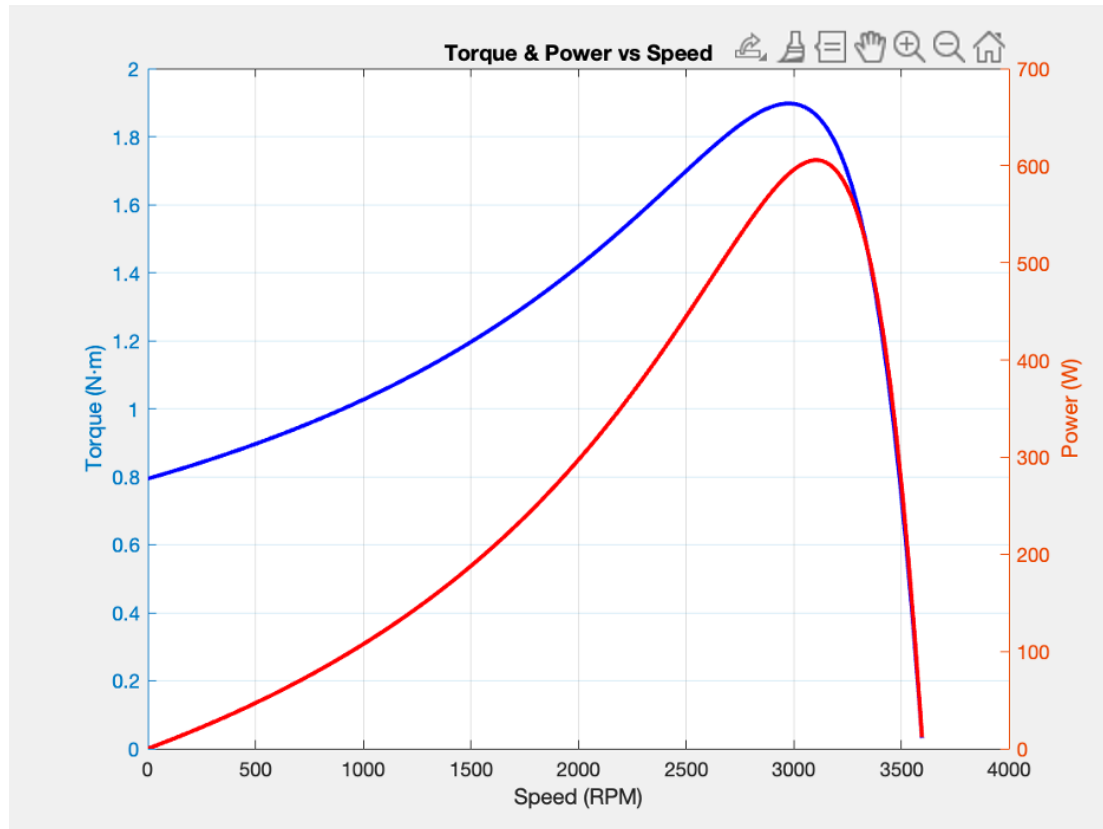
Simulating in MATLAB in combination with physical measurements and analytical methods of determining parameters. A model was able to be formed.



4. Simulation: verification and prediction

The behavior of this motor is modeled using a single phase equivalent circuit in MATLAB. This model incorporated all the parameters listed in Section 3.

The initial goal for this simulation is to verify and tweak parameters to see if the model can replicate the 1.5HP motor's design parameters of 1.1kW.



Cross referencing the MATLAB simulations, there is a discrepancy with the power. It is clearly under the specs of the motor. However, the motor is intended to be run at 230/460V, where the experiment used a 120V source with a variable frequency drive stepping to 3 phase, so more precise measurements need to be taken to find a more accurate max power at the designated maximum speed of 3450rpm.

5. Measurements

The first step in taking measurements was to find the voltage, current, and motor speed at different frequencies and different motor states. We used the Variable Frequency Drive to take measurements at 10, 20, 30, and 40 Hz.

The first set of measurements were taken with the motor in free spinning state. This process lacks a load which causes all current to go through the magnetization branch, this data will be used later to find the magnetization impedances.

Next was the stalled rotor, where we took an additional measurement: locked rotor torque. We utilized a wrench that clamped the rotor down and had that press into a scale recording torque. In the locked rotor stage, both slip and motor speed are zero.

3-PHASE VFD DRIVEN MOTOR CALCULATIONS				
MEASUREMENT	CALCULATED VALUES			
Frequency - [Hz]	10	20	30	40
Free Spinning Power - [VA]				
Locked Rotor Power - [VA]				
Torque - [Nm]				
Slip - [%]				
R2 - [Ohms]				
MEASUREMENT	MEASURED VALUES - FREE SPINNING			
Single Phase Voltage - [V]	120	120	120	120
Single Phase Current - [A]	0.9	1.2	1.6	1.9
Motor Speed - [RPM]	592	1196	1793	2387
L1-L2 Voltage - [V]	122.5	167.5	203.0	233.0
L1-L3 Voltage - [V]	122.5	167.5	203.0	233.0
L2-L3 Voltage - [V]	122.5	167.5	203.0	233.0
L1 Current - [A]	1.3	2.0	2.3	2.5
L2 Current - [A]	1.3	2.0	2.3	2.5
L3 Current - [A]	1.3	2.0	2.3	2.5
MEASUREMENT	MEASURED VALUES - STALLED ROTOR (I=8.25in)			
Single Phase Current - [A]	5.2	17.4	28.0	36.0
L1-L2 Voltage - [V]	105	133	140	143
L1-L3 Voltage - [V]	105	133	140	143
L2-L3 Voltage - [V]	105	133	140	143
Down Force - [g]	1570	2823	2917	2743

The next step was using these measurements to solve for the desired parameters. We needed to solve the DC resistance between two lines, the impedances of the magnetization branch, R_s , locked rotor torque, I_l , x .

Line to Line DC resistance:

- This measurement can be taken manually with a multimeter.
- Measured value is $R = 3\Omega$

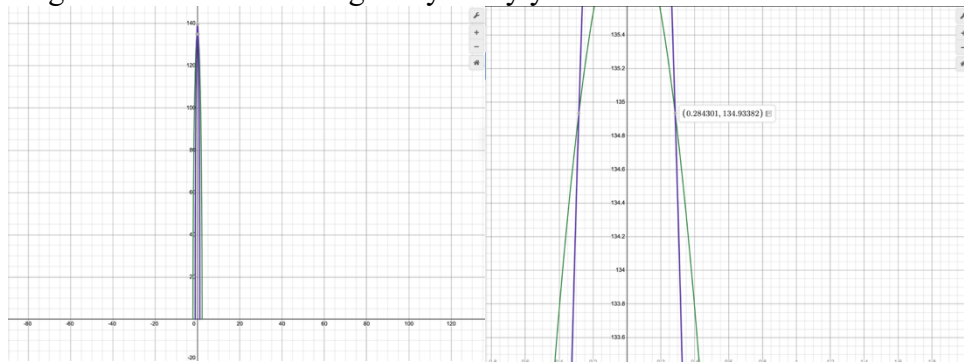
Magnetization Branch Impedances: ($Z_m = R_m + jX_m$)

- Using the free spinning measurements, only the magnetization branch is being utilized in the circuit, as there is no load.
- Utilizing the measurements of current and voltage through the magnetization branch, one has two unknowns.
- Using the system of equations:

- $Z_m = R_m + jX_m$
- $|Z_m| = \sqrt{R_m^2 + (2fL)^2}$
- $|V| = |I||Z_m|$
- $|V|^2 = R_m^2 + (2fL)^2$
- One can finally get this equation
 - $R_m = (|V||I|)^2 - (2fL)^2$
 - Unknowns: R_m and L
 - Now solving for knowns yields all values in the table below:

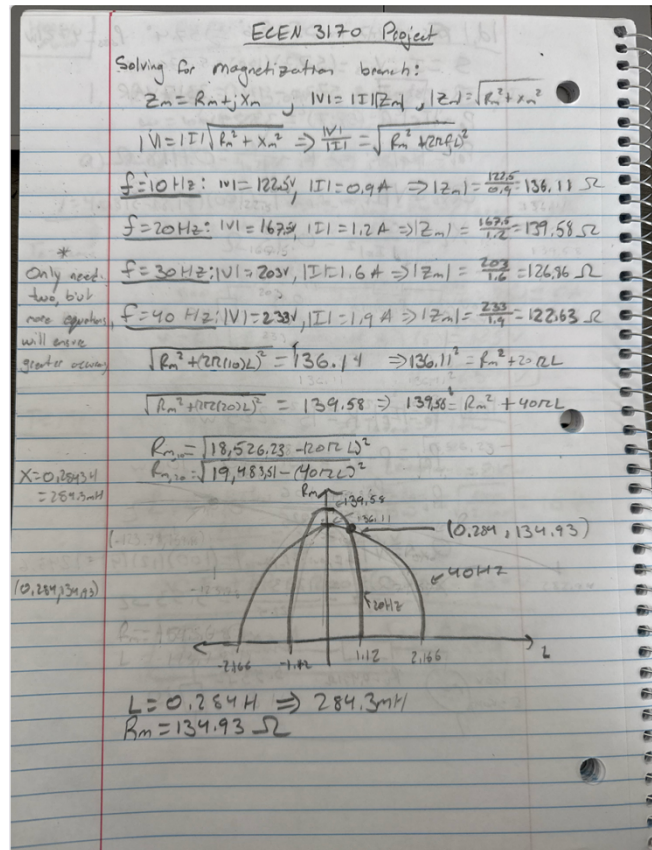
Frequency (Hz)	$ V $ (v)	$ I $ (A)	$ Z_m $ (Ω)
10	122.5	0.9	136.11
20	167.5	1.2	139.58
30	203	1.6	126.86
40	233	1.9	122.63

- Plotting these values or solving analytically yields:



(Plotted Frequencies at 10 and 20 Hz)

- $L = 284.3\text{mH}$
- $R_m = 134.93\Omega$



Locked Rotor Torque and Current:

- The objective of this section is to find the parameters R_2 , I_1 , and x
- Known Equations
 - $T_{LR} = |I_1|^2 s R_2$
 - $I = V Z_m + I_1$
 - $I_1 = V r_1 + R_2 + jx$
- Assumptions
 - Magnetization current is small, allowing the assumption $I \cong I_1$
- Finding R_2 :
 - Using physical measurements: Locked Rotor Torque is found, but needs to be converted from grams to Nm
 - Using the conversion $T = (\text{grams}) 9.81 \times 10^{-5}$
 - $r = 8.75 \text{ inches} \rightarrow 0.20955 \text{ m}$
 - The next parameter to solve for is synchronous speed (ω_s)
 - $s = 2f / \# \text{poles}$, $\# \text{poles} = 2$
 - Next, creating a table to handle the data to solve for R_2
 - Reworking the given equations creates: $R_2 = T_{LR} s / |I_1|^2$

f (Hz)	T_{LR} (Nm)	ω_s (rad/sec)	R_2 (Ω)
10	3.227	62.83	7.498
20	5.803	125.66	2.409

30	5.996	188.50	1.44
40	5.639	251.33	1.09

- Using these values and cross referencing them with our measured line-to-line voltage of 3Ω . Any value over 1.5 would not make sense.
- Using more accurate values of R_2 allows us to go back and find an accurate I_1 value, so we can find x
- Now we can find more accurate values for I_1 and Not approximate $I = I_1$ using
 - $I_1 = I - VZ_m$

f (Hz)	I_1 (A)
30	26.40
40	34.83

- Find X:
- Every parameter is found now to solve for x , so we can rework the equation
 - $I_1 = V / (r_1 + R_2 + jx)$
 - $x = V - I_1(r_1 + R_2) / I_1$

f (Hz)	$x (\Omega)$
30	3.8
40	2.61

Torque and Power Curves:



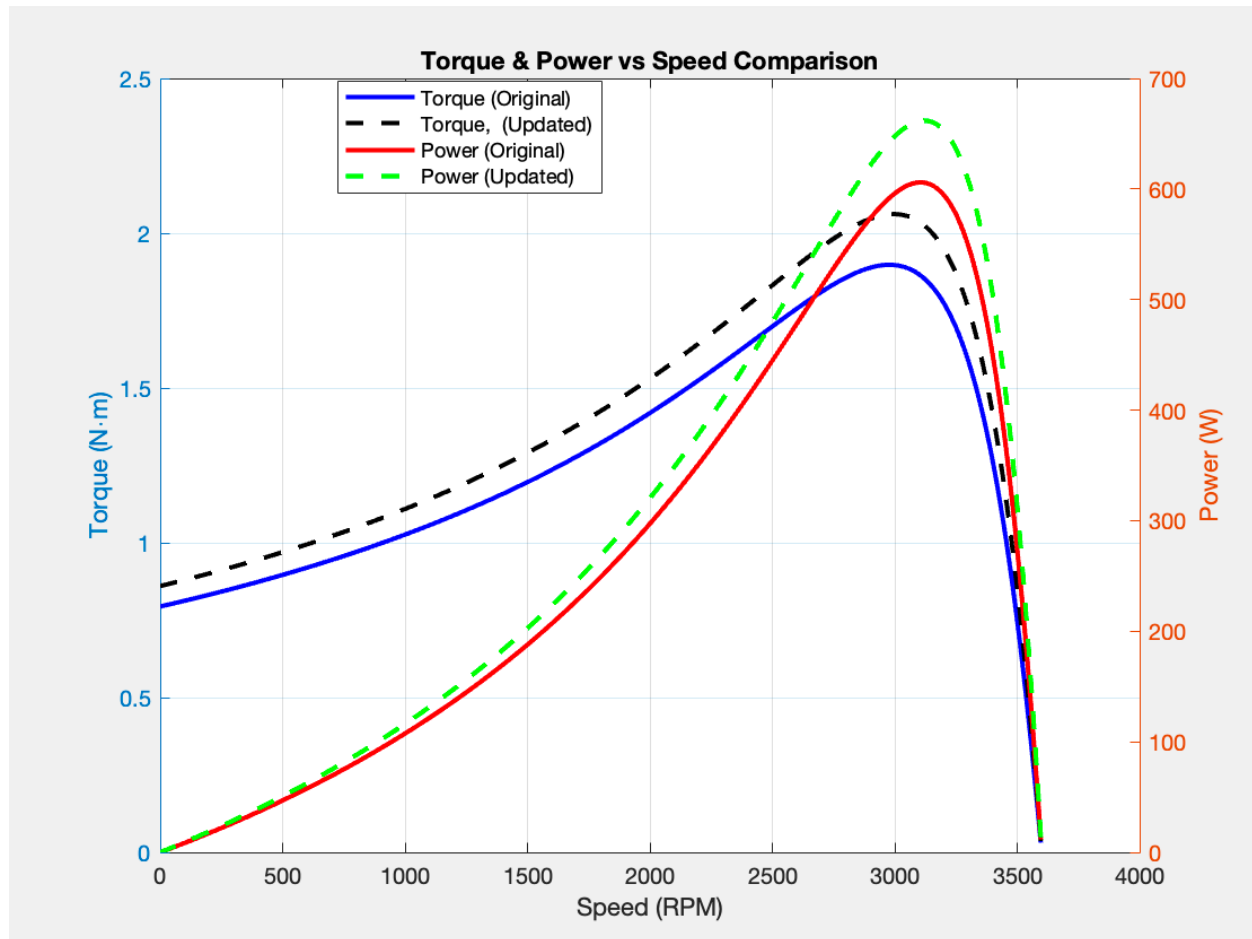
Using all the new measured parameters there is an increase in power and torque, which is closer to what the specs of the motor state.

6. Comparison of Measurements to Simulations

This section compares the initial model with the model derived from experimental and analytical measurements at 30 and 40 Hz. As frequency increased to 40 Hz. The measurement for R_2 was determined to be 1.09Ω , which fits in the bounds as $R=3=2(r_1+R_2)$. The reactance, x , was determined to be 2.61Ω determined from the locked rotor testing, and X_m was found from L to be 107.12Ω .

Changes in Performance:

The power output approaches the rated 1.1 kW maximum with the defined measurements. Even though the yielded power is less than the rating, it is likely due to other factors such as lower measurements at lower frequencies and a 120V source.



7. Conclusion

This project created a model and validated a single phase equivalent circuit model for a 1.5HP, three phase induction motor using measured parameters from both a no load (free spinning) and locked rotor tests, utilizing a Variable Frequency Drive to perform such measurements, ranging from frequencies of 10 to 40 Hz. The objective was to model the motor's performance across its range of speeds, allowing a comparison of the model to the specifications of the motor.

Parameter Identification:

Through the isolation of the magnetization branch during the free spinning tests, and additional measurements through the locked rotor tests, the following parameters were able to be identified:

- $R_m = 134.94 \, \Omega$
- $L = 284.3 \, \text{mH}$
- $R_2 = 1.09 \, \Omega$
- $X = 2.61 \, \Omega$

Power and Torque versus Speed Performance (MATLAB Derived)

The first MATLAB simulations based on physical measurements underpredicted the motor's specifications of 1.1kW. It is important to note that the motor's specifications were designed for 230V/460V, which is not what was used in the experimentation. Therefore, it is expected that our

measurements of power and torque will be less than what is stated for the motor. However, through rigorously solving for parameters, a more accurate model can be produced. By updating the model with all the solved parameters, selectively choosing the middle range of frequency (30,40 Hz) for even greater accuracy, a new model with power and torque curves greater than before was produced. Comparing the new model with the specifications of the motor, the peak values aligned closer with the maximum speed of 3450rpm and had closer values of power and torque to the specs.

Adjustments for the Future

To further increase the accuracy of this experiment, one should test the motor with its intended voltage, conduct a full load test in addition to the free spinning and locked rotor, and take more measurements up to the desired frequency of 60Hz. Taking measurements with the rated voltage will ensure that the magnetization branch matches the intention of the design specifications. Secondly, a full load test would allow one to have a middle group between the edge cases of no load and locked rotor. Lastly, extending our measurements up to the rated 60Hz frequency would allow a complete picture of the motor's performance under US conditions.

Appendix

Motor Specifications:

Group 4: Neill, Henry, Anusha, Brendan
M5.2 - M5.3, M6.5 slides 152-154
3-phase ½ HP induction motor
Can be operated at 208 and 480 V
Includes

- Measurement: current
- Modeling
- Tools: scope, tacho

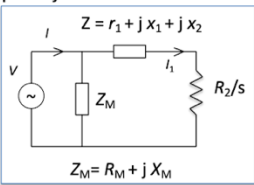
To be combined with a 3-phase variable frequency drive (VFD)
VFD Runs on 120V

Induction motor equivalent circuit

- Measure AC current and voltage at 10, 20, 30, 40 Hz
 - Locked rotor – measure torque
 - Free spinning – measure speed
- Measure power delivered to the VFD
- Use spec-sheet for speed at maximum power
- Extract R_M , X_M , r_1 , R_2
- Note that X_M changes with frequency

$$P_{mech,max} = \frac{V^2}{2(|Z + R_2| + r + R_2)}$$

$$s = \frac{R_2}{|Z + R_2| + R_2}$$





Brand	FAHKN5
Model Name	161332
Speed	3450 RPM
Voltage	460 Volts
Horsepower	1.5

About this item

- Three Phase General Motor: HP - 1.5, POLE - 2, FRAME - 56C, ENC -TEFC, HZ: 60Hz, Voltage: 230/460V, AMP: 4.6/2.3A, IP - 55, INS - F.
- High Working Efficiency: Compact structure and small size, electric motor can run at the maximum speed of 3450 rpm, high working efficiency; There are lots of heat sinks, so it has huge heat sinks.
- Fully Enclosed Fan: Electric motor with large fan at one end, even in wet and dusty conditions, cools the motor during long runs, thus reducing energy costs, reducing operating temperatures and increasing service life.
- Housing Material: Rolled steel shell. Adopt S1 continuous working system, safe and durable, long service life, Shaft diameter - 5/8, Shaft length - 1.88.
- Widely Application: Widely used in industrial and agricultural production, transportation, commercial and household appliances and other fields. For example, water pump, agricultural machinery and so on.

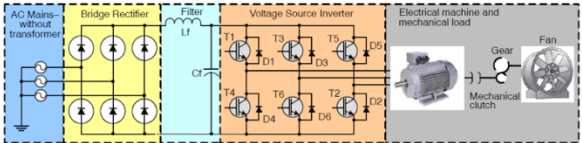


Figure 2. Common Topology of Motor Drive